Factor Analysis

Chapter 24

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Agenda

What is factor analysis?
The factor model
Conducting factor analysis
Full worked example: customer satisfaction
Further example: screening job applicants
More real research
SPSS and summary

Learning objectives

After this session you should be able to:

1. Explain factor analysis as **data reduction and summarisation**, and how it differs from ANOVA, regression and discriminant analysis.
2. Run through the full procedure: formulate the problem, build the correlation matrix, choose a method, decide the number of factors, rotate and interpret.
3. Distinguish **principal components analysis** from **common factor analysis**.
4. Compute and use **factor scores**, or pick **surrogate variables** instead.
5. Judge model fit with **reproduced correlations** and residuals.
6. Run and read a factor analysis in SPSS.

Hidden constructs we cannot observe directly

Key idea

Factors are hidden (**latent**) constructs that are correlated with things we *can* observe. Factor analysis lets us quantify what we do not see directly.

- Style, extraversion, price-consciousness: none is tangible, and all are "fuzzy".
- But each leaves traces in observable variables:
  - *Style* $\rightarrow$ brands bought, amount spent on clothing.
  - *Extraversion* $\rightarrow$ number of parties attended, social diary.
  - *Price-consciousness* $\rightarrow$ seeking discounts, "trading down".
- Factor analysis exploits the **correlations** among the observed variables to recover the few constructs behind them.

Basic concept

Factor analysis

A class of procedures used mainly for **data reduction and summarisation**. A **factor** is an underlying dimension that explains the correlations among a set of variables.

- In ANOVA, regression and discriminant analysis one variable is the *dependent* variable.
- Factor analysis makes **no** such distinction: it is an **interdependence technique**, examining the whole set of interrelationships at once.
- Data reduction is *column-wise*: reduce k variables to $m < k$ factors.
  (Cluster analysis does the row-wise counterpart: group similar *cases*.)

	Variable 1	$\cdots$	Variable k
Observation 1	X_{11}	$\cdots$	X_{1k}
$\vdots$	$\vdots$		$\vdots$
Observation n	X_{n1}	$\cdots$	X_{nk}

Example: why do people buy shampoo?

A questionnaire asks $k = 20$ questions about shampoo. Choice is probably driven by a handful of underlying constructs:

- value for money
- desired hair effect (shine, volume, anti-dandruff)
- suitability for hair type (damaged, dyed, dry)
- usage attributes (easy rinsing, no irritation)
- bottle design (shape, colour, recyclable)

- Goal: recover these ≈ 5 factors from the 20 observed variables.
- Price and perceived-quality questions should be correlated, so they would form a *value-for-money* factor. Note that value for money is not the same as "cheap".
- Because factors are fuzzy, they must be **easy to interpret and name**, even if the naming is somewhat subjective.
- Once identified, each respondent can be **scored** on each factor for further analysis.

Real research

Real research: Personal alarms

Women rated eight personal alarms on 15 statements (e.g. "feels comfortable in the hand", "would fit easily into a handbag", "difficult for an attacker to switch off", "looks as if it would give off a very loud noise", "I would be embarrassed to carry it").

Question: could the 15 variables be reduced to fewer derived variables without losing too much information?

Result: four factors: (1) *size*, from small to large; (2) *appearance*; (3) *robustness*; (4) *hand feel*.

Real research: Myers–Briggs personality types

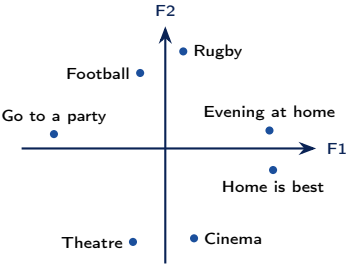
Four "factors" (personality traits), each with two poles: E/I, N/S, F/T, J/P. Questions such as "Do you enjoy group activities?" (E vs I) or "Are you a very organised person?" (J vs P) produce factor scores, which then classify people into $2^4 = 16$ types. Managers can use the types to build balanced teams. The constructs are fuzzy (what exactly is a "feeler"?), which is typical of factors.

Three uses of factor analysis

1. **Identify underlying dimensions** that explain the correlations, e.g. psychographic factors behind lifestyle statements.
2. **Create a smaller set of uncorrelated variables** (factor scores) to replace the correlated originals in regression or discriminant analysis. This removes **multicollinearity**.
3. **Select a few salient original variables** (e.g. "Home is the best place" and "Football") to use in place of all of them.

All three uses are exploratory, so the method is also called **exploratory factor analysis (EFA)**.

Figure 1. Factors behind psychographics (illustrative)



F1: homebody vs socialite. F2: sports vs cinema/theatre.

Applications in marketing research

- **Market segmentation**: find the variables to group customers on. New-car buyers may weigh economy, convenience, performance, comfort and luxury, which gives five segments.
- **Product research**: find the brand attributes that drive choice. Toothpaste: cavity protection, whiteness, taste, fresh breath, price.
- **Advertising**: understand media habits. Frozen-food buyers may watch horror films, play electronic games and listen to rock music.
- **Pricing**: profile price-sensitive consumers, who may be methodical, economy-minded and home-centred.

The factor model

Work with **standardised** variables: $X'_j = \dfrac{X_j - \bar{X}_j}{S_j}$, so each has mean 0 and variance 1.

Each standardised variable is a linear combination of m **common factors** plus a **unique (specific) factor**:

$$X'_j = \alpha_{1j}F_1 + \alpha_{2j}F_2 + \cdots + \alpha_{mj}F_m + V_j, \qquad j = 1, \ldots, k$$

- α_{ij} = **factor loadings**: the correlations between variable and factor.
- V_j = variation specific to X'_j, which the common factors cannot explain.

The factors, which we do not observe, are estimated as linear combinations of the observed variables:

$$F_i = W_{i1}X'_1 + W_{i2}X'_2 + \cdots + W_{ik}X'_k$$

The W are **factor score coefficients**. They are chosen so that F_1 explains the most variance, F_2 the most of what remains while being uncorrelated with F_1, and so on.

Assumptions and communality

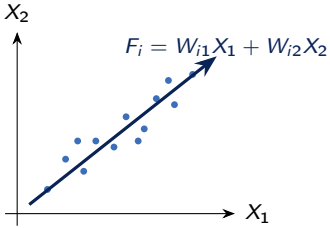
Assumptions

- $E(F_i) = 0, \ \ \text{Var}(F_i) = 1$
- Factors uncorrelated: $\text{Cov}(F_i, F_j) = 0, \ i \neq j$
- $E(V_j) = 0$; unique terms uncorrelated with each other
- Factors and unique terms uncorrelated: $\text{Cov}(F_i, V_j) = 0$

Variance decomposition

$$1 = \text{Var}(X_j') = \underbrace{\alpha_{1j}^2 + \cdots + \alpha_{mj}^2}_{\text{communality}} + \underbrace{\text{Var}(V_j)}_{\text{specific variance}}$$

Figure 2. Factor analysis with two variables



We want communalities close to 1.

Win and cost: uncorrelated factors remove multicollinearity and simplify the model (Occam's razor), but factors may be hard to label and some information is lost.

Key statistics (1/2)

Bartlett's test of sphericity Tests H_0: the variables are uncorrelated in the population, i.e. the population correlation matrix is the identity matrix.

Communality The share of a variable's variance it has in common with the other variables, i.e. the share explained by the common factors.

Correlation matrix Lower triangle of the simple correlations r between all pairs of variables.

Eigenvalue Total variance explained by a factor.

Factor loadings Simple correlations between the variables and the factors.

Factor loading plot Variables plotted with their factor loadings as coordinates.

Factor matrix The loadings of all variables on all extracted factors.

Key statistics (2/2)

Factor scores Composite scores estimated for each respondent on the derived factors.

Factor score coefficient matrix The weights used to combine the standardised variables into factor scores.

KMO measure of sampling adequacy Compares the observed correlations with the partial correlations. Values of 0.5–1.0 mean FA is appropriate; values below 0.5 mean it may not be.

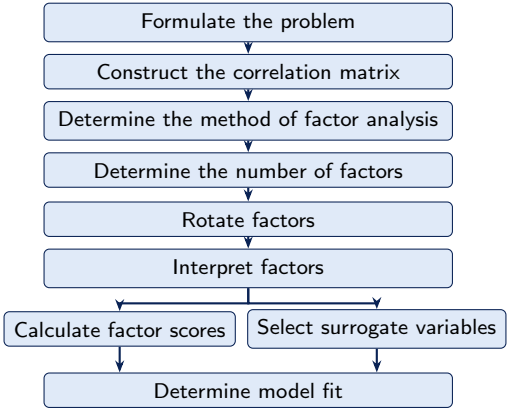
Percentage of variance Share of the total variance attributed to each factor.

Residuals Observed correlations minus reproduced correlations (reproduced from the factor matrix).

Scree plot Eigenvalues plotted against the number of factors, in order of extraction.

The procedure

Figure 3. Steps in conducting factor analysis



Step 1 – Formulate the problem

- State the objectives; choose variables from past research, theory and the researcher's judgement.
- Variables should be measured on an **interval or ratio** scale.
- Sample size: as a rough guide, at least **4–5 observations per variable**. If you have fewer, interpret the results with caution.

Running example: toothpaste benefits ($n = 30$, street interviews)

Agreement on a 7-point scale (1 = strongly disagree, 7 = strongly agree):

V1 It is important to buy a toothpaste that prevents cavities.
V2 I like a toothpaste that gives shiny teeth.
V3 A toothpaste should strengthen your gums.
V4 I prefer a toothpaste that freshens breath.
V5 Prevention of tooth decay is *not* an important benefit offered by a toothpaste.
V6 The most important consideration in buying a toothpaste is attractive teeth.

Table 24.1 – Toothpaste attribute ratings

ID	V1	V2	V3	V4	V5	V6
1	7	3	6	4	2	4
2	1	3	2	4	5	4
3	6	2	7	4	1	3
4	4	5	4	6	2	5
5	1	2	2	3	6	2
6	6	3	6	4	2	4
7	5	3	6	3	4	3
8	6	4	7	4	1	4
9	3	4	2	3	6	3
10	2	6	2	6	7	6

ID	V1	V2	V3	V4	V5	V6
11	6	4	7	3	2	3
12	2	3	1	4	5	4
13	7	2	6	4	1	3
14	4	6	4	5	3	6
15	1	3	2	2	6	4
16	6	4	6	3	3	4
17	5	3	6	3	3	4
18	7	3	7	4	1	4
19	2	4	3	3	6	3
20	3	5	3	6	4	6

ID	V1	V2	V3	V4	V5	V6
21	1	3	2	3	5	3
22	5	4	5	4	2	4
23	2	2	1	5	4	4
24	4	6	4	6	4	7
25	6	5	4	2	1	4
26	3	5	4	6	4	7
27	4	4	7	2	2	5
28	3	7	2	6	4	3
29	4	6	3	7	2	7
30	2	3	2	4	7	2

Small n for illustration only; in practice factor analysis uses much larger samples.

Step 2 – Construct the correlation matrix

Table 24.2 – Correlation matrix

	V1	V2	V3	V4	V5	V6
V1	1.00					
V2	−0.053	1.00				
V3	**0.873**	−0.155	1.00			
V4	−0.086	**0.572**	−0.248	1.00		
V5	**−0.858**	0.020	**−0.778**	−0.007	1.00	
V6	0.004	**0.640**	−0.018	**0.640**	−0.136	1.00

- For FA to make sense the variables **must be correlated**. If all correlations are small, FA is not appropriate.
- Two clear clusters:
  - V1, V3, V5 (cavities, gums, decay)
  - V2, V4, V6 (shine, breath, attractiveness)
- Highly correlated variables should load on the same factor(s).

Formal appropriateness tests

Bartlett: H_0 : P = I (χ^2 transformation of the determinant of the correlation matrix; large values reject H_0). **KMO:** compares observed correlations with partial correlations; > 0.5 is desirable.
Toothpaste: $\chi^2 = 111.314$, $df = 15$, $p < 0.001$; KMO $= 0.660$. $\Rightarrow$ FA is appropriate.

Step 3 – Determine the method

The two basic approaches differ in what goes on the diagonal of the correlation matrix:

Principal components analysis (PCA)

- Uses the **total** variance; 1s on the diagonal.
- Goal: the *minimum number of factors* that account for the *maximum variance*, to use in later multivariate analysis.
- The factors are called **principal components**.

Common factor analysis

- Uses only the **common** variance; communalities on the diagonal.
- Goal: identify the *underlying dimensions*, when the common variance is what matters.
- Also called **principal axis factoring**.

Other methods (unweighted/generalised least squares, maximum likelihood, alpha, image factoring) are more complex. Good practice: try more than one method as a *sensitivity analysis*. If the conclusions agree, the results are robust.

Toothpaste: PCA results (Table 24.3)

Communalities

Var	Initial	Extraction
V1	1.000	0.926
V2	1.000	0.723
V3	1.000	0.894
V4	1.000	0.739
V5	1.000	0.878
V6	1.000	0.790

Initial eigenvalues

Factor	Eigenvalue	% variance	Cum. %
1	**2.731**	45.520	45.520
2	**2.218**	36.969	**82.488**
3	0.442	7.360	89.848
4	0.341	5.688	95.536
5	0.183	3.044	98.580
6	0.085	1.420	100.000

- Initial communalities are 1 because PCA puts 1s on the diagonal. The "Extraction" column shows how much is explained once only the retained factors are kept.
- Total variance $= 6$ (one per standardised variable). Factor 1: $2.731/6 = 45.52\%$.
- In PCA, the "extraction sums of squared loadings" equal the initial eigenvalues of the retained factors.

Step 4 – Determine the number of factors

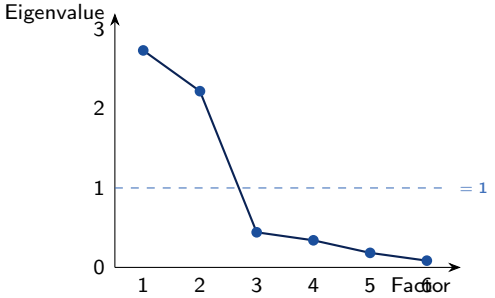
You can extract up to k factors, but that gives no parsimony. Six approaches:

1. **A priori**: prior knowledge says how many factors to expect.
2. **Eigenvalues > 1** (Kaiser): keep factors that explain more than one standardised variable does. Conservative when $k < 20$.
3. **Scree plot**: look for the break between the steep slope and the flat "scree". Often gives one or a few more factors than the eigenvalue rule.
4. **Percentage of variance**: keep factors until the cumulative variance is satisfactory, usually **at least 60%**.
5. **Split-half reliability**: factor each half of the sample and keep factors whose loadings agree across the halves.
6. **Significance tests**: keep only significant eigenvalues. With $n > 200$ many trivial factors become significant.

Ideally, several criteria point to the same number of factors.

Toothpaste: how many factors?

Figure 4. Scree plot – toothpaste data



- **Eigenvalue** > 1: 2 factors.
- **A priori**: toothpaste is bought for two main reasons.
- **Scree**: distinct break at factor 3.
- **Cumulative variance**: 82.49% with 2 factors; the third adds little.
- **Split-half**: also supports 2.

Decision

Retain **two factors**.

Step 5 – Rotate factors

- The unrotated **factor (pattern) matrix** seldom gives interpretable factors, because many variables load on several factors.
- Once m is fixed the solution is not unique: rotating the axes leaves total variance explained unchanged and only redistributes it.
- Goal: each variable loads **high on one factor and low on the others**.
- **Orthogonal** rotation keeps the axes at 90°, so the factors stay uncorrelated:
  - **Varimax**: fewest variables with high loadings per factor (most common)
  - Quartimax: fewest factors needed per variable; Equamax: a mix
- **Oblique** rotation lets factors correlate; use it when they are likely strongly correlated in the population.

Figure 5. High loadings before and after rotation

(a) Before

Var	F1	F2
1	X	
2	X	X
3	X	
4	X	X
5	X	X
6		X

(b) After

Var	F1	F2
1	X	
2		X
3	X	
4		X
5	X	
6		X

Toothpaste: unrotated vs varimax-rotated

Factor matrix (unrotated)

	F1	F2
V1	**0.928**	0.253
V2	−0.301	**0.795**
V3	**0.936**	0.131
V4	−0.342	**0.789**
V5	**−0.869**	−0.351
V6	−0.177	**0.871**

Rotated factor matrix (varimax)

	F1	F2
V1	**0.962**	−0.027
V2	−0.057	**0.848**
V3	**0.934**	−0.146
V4	−0.098	**0.854**
V5	**−0.933**	−0.084
V6	0.083	**0.885**

Rotation sums of sq. loadings	Eigenvalue	% var.	Cum. %
Factor 1	2.688	44.802	44.802
Factor 2	2.261	37.687	82.488

Before rotation, five variables have sizeable loadings on F1 (V2, V4 and V6 load −0.18 to −0.34 on it). After rotation only V1, V3 and V5 load on F1, and V2, V4 and V6 load on F2; no variable loads highly on both. Total variance explained is still 82.49%.

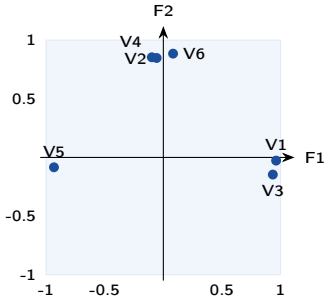
Step 6 – Interpret factors

- Name each factor after the variables that load highly on it.
- On the loading plot, variables at the end of an axis define that factor. Variables near the origin load on neither factor; variables away from both axes relate to both.
- If a factor cannot be clearly defined, label it *undefined* or *general*.

Toothpaste

F1 = health benefits: V1 (+), V3 (+), V5 (−).
A negative loading on a negatively worded item (V5) has a positive meaning.
F2 = social benefits: V2, V4, V6.

Figure 6. Factor loading plot (rotated) – toothpaste



Step 7a – Calculate factor scores

Factor score coefficients

	F1	F2
V1	0.358	0.011
V2	−0.001	0.375
V3	0.345	−0.043
V4	−0.017	0.377
V5	−0.350	−0.059
V6	0.052	0.395

$F_i = W_{i1}X'_1 + \cdots + W_{ik}X'_k$: multiply the *standardised* values by the coefficients.

- Exact scores exist only for PCA, and they are uncorrelated.
- In common FA the scores are estimates and may be correlated.

Worked example: participant 1 (V = 7, 3, 6, 4, 2, 4)

Standardised: $X' = (1.55, -0.66, 0.92, -0.07, -0.79, -0.12)$

$$\text{Health} = 0.358(1.55) - 0.001(-0.66) + 0.345(0.92) - 0.017(-0.07) - 0.350(-0.79) + 0.052(-0.12) \approx 1.14$$

Social ≈ -0.30

Participant 10 (2, 6, 2, 6, 7, 6): Health ≈ -1.30, Social ≈ 1.54.

Means/SDs from Table 24.1, e.g. $\bar{V}_1 = 3.93$, $S_1 = 1.98$. Scores can replace the original variables in regression or discriminant analysis.

Step 7b – Select surrogate variables

Surrogate variable

One of the original variables, chosen to stand in for a factor in later analysis. The results can then be interpreted in terms of an observed variable rather than a synthetic score.

- Default rule: for each factor, pick the variable with the **highest loading**.
- This works well when one loading is clearly higher than the others. When several are close, decide on **theory and measurement quality**: a slightly lower loading measured more precisely may be the better choice.
- Toothpaste, F1: V1 (0.962), V3 (0.934) and V5 (-0.933) are all close. V1 is the default; pick V5 if prior knowledge says decay prevention is the key benefit.
- Toothpaste, F2: V2, V4 and V6 are all comparable. Pick V6 if attractive teeth is the key social benefit.

Step 8 – Determine model fit

- FA assumes the observed correlations come from the common factors, so the factor matrix should **reproduce** them.
- Residual $= r_{ij} - \tilde{r}_{ij}$ (observed minus reproduced).
- Many large residuals mean poor fit, and the model should be reconsidered.
- Toothpaste: only **5** residuals exceed 0.05, so the fit is acceptable.

Reproduced correlation matrix (PCA)

	V1	V2	V3	V4	V5	V6
V1	**0.926**	0.024	−0.029	0.031	0.038	**−0.053**
V2	−0.078	**0.723**	0.022	**−0.158**	0.038	**−0.105**
V3	0.902	−0.177	**0.894**	−0.031	**0.081**	0.033
V4	−0.117	0.730	−0.217	**0.739**	−0.027	**−0.107**
V5	−0.895	−0.018	0.859	0.020	**0.878**	0.016
V6	0.057	−0.746	−0.051	0.748	−0.152	**0.790**

Lower triangle: reproduced correlations (as printed in Table 24.3). Diagonal: communalities. Upper triangle: residuals; those with $|\cdot| > 0.05$ are in blue.

Common factor analysis on the same data (Table 24.4)

	Communality		Rotated	
	Initial	Extr.	F1	F2
V1	0.859	0.928	**0.963**	−0.030
V2	0.480	0.562	−0.054	**0.747**
V3	0.814	0.836	**0.902**	−0.150
V4	0.543	0.600	−0.090	**0.769**
V5	0.763	0.789	**−0.885**	−0.079
V6	0.587	0.723	0.075	**0.847**

Extraction	F1	F2
Eigenvalue	2.570	1.868
% variance	42.84	31.13
Cumulative %	42.84	73.96

- The initial communalities are now **below 1**, because communalities replace the 1s on the diagonal.
- Bartlett and KMO are unchanged; two factors are extracted again.
- Each factor explains a little less variance than in PCA (73.96% vs 82.49%), because only the common variance is modelled.
- The rotated pattern is the same, so the interpretation is the same: *health* and *social* benefits.
- Sometimes CFA and PCA assign variables to different factors. With more than 15 variables they usually agree. PCA is less prone to misinterpretation and is recommended for non-experts.

Case: retailer satisfaction survey

A retailer surveyed $n = 150$ customers on five attributes, each on a 5-point Likert scale (1 = strongly negative, 5 = strongly positive):

- X_1 price satisfaction
- X_2 product range satisfaction
- X_3 packaging satisfaction
- X_4 customer service satisfaction
- X_5 product quality satisfaction

Goal: find the underlying dimensions that explain the correlations among these five responses.

SPSS settings used

Principal components extraction; eigenvalues > 1; correlation coefficients, significance levels, reproduced correlations, KMO and Bartlett; scree plot; **varimax** rotation with loading plot; save factor scores and display the score coefficient matrix.

Is factor analysis appropriate?

Correlation matrix (one-tailed tests, $t = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$). Significant pairs:

- X_2–X_4, X_2–X_5, X_4–X_5: $p < 0.001$
- X_3–X_4 ($p = 0.006$), X_3–X_5 ($p = 0.008$)
- X_1–X_5: only at 10% ($p = 0.084$)

Bartlett's test ($k = 5$): $\chi^2 = 132.152$ on 10 df, $p < 0.001 \Rightarrow$ reject H_0 : P = I.

KMO: should be at least 0.5 (the higher the better, up to 1).

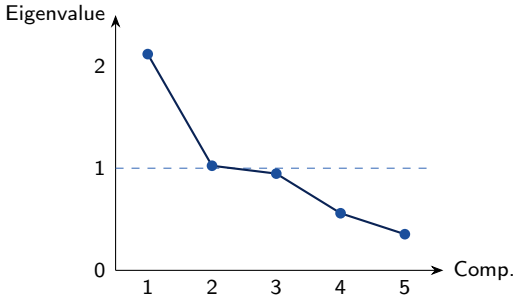
Communalities

X_1 0.925 X_2 0.705 X_3 **0.196** X_4 0.672 X_5 0.643

All are "high" (above about 0.6) *except packaging satisfaction*, which shares little variance with the others.

How many factors?

Figure 7. Scree plot – satisfaction survey



- Eigenvalues: 2.117, 1.024, 0.947, 0.559, 0.354, so **2** exceed 1.
- Variance: $42.340\% + 20.471\% = 62.810\%$, above the 60% benchmark.
- Price of the reduction: $100\% - 62.81\% = 37.19\%$ of the variance is discarded.
- **Caveat**: components 2 and 3 are almost equal (1.024 vs 0.947). The scree is hard to read and the cutoff is close to arbitrary.
- Is $5 \to 2$ worth the information loss? It depends, mainly on whether the factors can be given meaningful names.

Rotated loadings and naming

High loadings, unrotated vs varimax

	Unrotated		Rotated	
	F1	F2	F1	F2
X_1 price		0.960		**0.959**
X_2 range	0.839		**0.838**	
X_3 packaging	–	–	–	–
X_4 service	0.817		**0.816**	
X_5 quality	0.787		**0.788**	

Only the high loadings are shown. X_3 (communality 0.196) loads highly on neither factor.

- Here the unrotated solution is already clear; rotation confirms it. In other data sets rotation makes a much bigger difference (see the next example).
- **Factor 1**: range, service and quality, which together describe the experience with products and service. Name: **customer experience**.
- **Factor 2**: price only. Name: **price**.
- On the loading plot (axes run from -1 to $+1$), X_2, X_4, X_5 sit at the right end of F1, X_1 at the top of F2, and X_3 near neither.

Computing the factor scores

Component score coefficients give:

$$\text{Experience} = F_1 = -0.036X_1' + 0.396X_2' + 0.165X_3' + 0.386X_4' + 0.372X_5'$$
$$\text{Price} = F_2 = \quad 0.937X_1' + 0.042X_2' + 0.265X_3' + 0.067X_4' - 0.147X_5'$$

with $X_1' = \frac{X_1 - 1.91}{0.822}$, $X_2' = \frac{X_2 - 3.25}{1.321}$, $X_3' = \frac{X_3 - 3.10}{1.384}$, $X_4' = \frac{X_4 - 3.27}{1.339}$, $X_5' = \frac{X_5 - 3.19}{1.402}$.

Two customers

	Raw $(X_1, \ldots, X_5)$	Standardised X'	F_1	F_2
Case 4	(2, 2, 2, 1, 3)	(0.114, −0.944, −0.795, −1.692, −0.133)	−1.211	−0.238
Case 10	(2, 4, 4, 5, 5)	(0.114, 0.570, 0.650, 1.294, 1.294)	1.309	0.199

Same price satisfaction, very different service satisfaction (1 vs 5), so the experience scores sit at opposite ends.

- The scores can serve as satisfaction indices. After varimax they are uncorrelated (the correlation matrix of the scores is the identity), so there is no multicollinearity in later regressions.
- Small discrepancies in hand calculations are due to rounding.

Surrogates and residual analysis

Surrogates: X_2 range (0.838) for F1; X_1 price (0.959) for F2.

Fit: reproduced diagonal = communalities. Example: $r_{12} = 0.021$, $\tilde{r}_{12} = -0.017$, so the residual is 0.038.

7 of 10 residuals exceed 0.05:

$$-0.216,\ 0.086,\ -0.223,\ -0.092,\ -0.117,\ -0.098,\ -0.202$$

- The ± 0.05 cutoff is strict and has an "edge-of-the-cliff" problem: 0.051 and 1.999 are both "bad".
- It is better to look at all the residuals (e.g. plot them). None is far from 0, so the fit is acceptable overall.

Figure 8. Classifying residuals (SPSS rule)



(good zone drawn wider than scale)

Residuals range from -2 to $+2$, since correlations lie in $[-1, 1]$. A perfect residual is 0, i.e. $r_{ij} = \tilde{r}_{ij}$.

Case: Righthome estate agency

Righthome, a new estate-agency chain, needs assertive sales negotiators. Its 48 applicants rated themselves from 0 (not at all) to 10 (completely) on $k = 11$ traits:

ambition X_1, drive X_2, experience X_3, grasp X_4, honesty X_5, likeability X_6,
lucidity X_7, potential X_8, salesmanship X_9, self-confidence X_{10}, suitability X_{11}

Appropriateness

Many strong, significant correlations; the only negative one is experience–honesty.
KMO $= 0.794$.
Bartlett: $\chi^2 = 510.313$, $df = 55$, $p < 0.001$.
All communalities > 0.75.

Number of factors

Three eigenvalues > 1; the scree flattens from factor 4.
$59.539 + 15.150 + 10.829 = 85.518\%$
Information lost: only 14.48% for an $11 \rightarrow 3$ reduction.

Rotation earns its keep

Unrotated: F1 is clear ($X_1, X_2, X_4, X_7, X_8, X_9, X_{10}$; loadings 0.83–0.92). F2 and F3 are *ambiguous*: X_3, "possibly" X_5 and X_{11} on F2; X_6 on F3.

Varimax-rotated high loadings:

Factor	Variables (loading)
F1	ambition .923, self-conf. .921, salesm. .901, lucidity .861, grasp .805, drive .790, potential .736
F2	experience .918, suitability .850
F3	honesty .876, likeability .874

Honesty (X_5) moved from the ambiguous F2 to F3.

Naming (subjective)

F1 = **assertiveness** F2 = **expertise** F3 = **character**

Surrogates would be ambition, experience and honesty. Because the purpose here is to measure assertiveness, the named factors are more useful than the surrogates.

- With 3 factors the loading plot is 3-D; with 4 or more factors no plot is possible, so read the rotated matrix instead.
- Residuals: 10 of 55 (18%) exceed 0.05, none extreme, so the fit is good.

Using the scores to short-list

$$\text{Assertiveness} = 0.245X_1' + 0.142X_2' - 0.134X_3' + \cdots - 0.059X_{11}'$$

(similar equations give expertise and character).

- Higher assertiveness score $\Rightarrow$ more assertive, the key trait for a negotiator.
- Higher expertise score $\Rightarrow$ more experienced and suitable (a proven track record).
- Higher character score $\Rightarrow$ more honest and likeable, which helps build rapport with buyers and sellers.

Managerial use

Managers weight the three factors by importance (is assertiveness worth more than character?) and short-list the applicants closest to the ideal profile. The scores are a **screening** tool only. Interviews still decide, and self-ratings assume honest answers.

Real research: trade promotions

Real research: Manufacturer trade-promotion components

Goal: inventory the manufacturer-controlled trade-promotion variables and test their link with retailers' promotion support.

FA: PCA with varimax reduced 30 variables to **8 factors** (eigenvalues > 1) explaining 62% of the variance. A variable was assigned to a factor if it loaded ≥ 0.40; cross-loaders went to their highest loading; one variable loaded on no factor.

F1 Item importance (16.3%)
F2 Promotion elasticity (9.3%)
F3 Manufacturer brand support (8.2%)
F4 Manufacturer reputation (7.3%)
F5 Promotion wearout (6.4%)
F6 Sales velocity (5.4%)
F7 Item profitability (4.5%)
F8 Incentive amount (4.2%)

Follow-up: the 8 *factor scores* were the predictors in a stepwise discriminant analysis of the buyer's rating (low/medium/high support), $n = 564$. All 8 entered, 65% of cases were correctly classified, and the most important influencers were item importance, then item profitability, then incentive amount.

This is use #2 in practice: factor scores replace correlated predictors in a later analysis.

Real research: rebates, Beetles and ethics

Real research: "Common" rebate perceptions

24 Likert items, $n = 303$. Common FA; the first solution lacked simple structure, so low-loading items were dropped and FA rerun, giving 3 factors (loadings 0.527–0.744): **Faith** in the rebate system; **Efforts** needed to redeem; **Motives** of manufacturers.
Implication (e.g. AT&T): minimise redemption effort, build faith, show honest motives.

Real research: Driving nuts for Beetles

Across countries, PCA of reasons for owning a VW Beetle consistently gives one factor: **fanatical loyalty**.

Real research: Unethical marketing research practices

420 marketing professionals, 11 items. PCA + varimax: *extent of ethical problems* (46%, $\alpha = 0.87$) and *top-management action on ethics* (11%, $\alpha = 0.75$). In a regression they were the two best predictors of unethical practices.

Running factor analysis in SPSS

Analyze > **Dimension Reduction** > **Factor...**

1. Move the variables (e.g. V1–V6) into VARIABLES. There is no dependent variable.
2. DESCRIPTIVES: Initial solution; Coefficients, Significance levels, **KMO and Bartlett's test**, **Reproduced**.
3. EXTRACTION: Method = **Principal components** (or **Principal axis factoring** for common FA); Analyze the correlation matrix; Based on eigenvalue > 1; Display the unrotated solution and **Scree plot**.
4. ROTATION: **Varimax**; Rotated solution; Loading plot(s).
5. SCORES: Save as variables (adds FAC1_1, FAC2_1, ... to the data); Display factor score coefficient matrix.
6. OK.

What to read in the output

Correlation matrix → KMO/Bartlett → communalities → total variance explained and scree → rotated component matrix and loading plot → score coefficients → reproduced correlations and residuals.

Summary

- Factor analysis (EFA) reduces and summarises data. Each variable is a linear combination of underlying factors, and each factor is a linear combination of the variables.
- Choose interval/ratio variables based on theory and past research. Check that they are correlated: correlation matrix, Bartlett, KMO > 0.5.
- **PCA** (total variance) finds the fewest factors for later analysis. **Common FA / principal axis** (common variance) finds the underlying dimensions.
- Choose the number of factors from: a priori knowledge, eigenvalues > 1, the scree plot, $\geq 60\%$ of variance, split-half reliability, significance tests.
- **Rotate** (usually varimax, which is orthogonal; oblique if the factors are correlated) and interpret from the rotated matrix.
- Use **factor scores** (uncorrelated in PCA) or **surrogate variables** in later analysis.
- Check **fit** with the reproduced correlations and residuals, judged as a whole rather than by a strict cutoff.

Key vocabulary

- Bartlett's test of sphericity
- Common factor analysis
- Communality
- Correlation matrix
- Data reduction
- Eigenvalue
- Factor
- Factor loading (plot)
- Factor matrix
- Factor rotation
- Factor score
- Factor score coefficient
- Interdependence technique
- KMO statistic
- Oblique rotation
- Orthogonal rotation
- Principal components
- Reproduced correlation
- Residual
- Scree plot
- Surrogate variable
- Varimax

Discussion questions

1. How is factor analysis different from regression and discriminant analysis?
2. A clothing retailer wants to measure *fashion consciousness*. Is it directly observable? Propose observed variables that should correlate highly with it.
3. Should the input variables be correlated or uncorrelated? Why?
4. What does Bartlett's test actually test? What is a "good" KMO value?
5. In the satisfaction example, components 2 and 3 have eigenvalues 1.024 and 0.947. How would you defend or challenge retaining two factors?
6. Explain, at a high level, what rotation does and why it does not change the total variance explained.
7. When would you prefer a surrogate variable to a factor score? Use the toothpaste example.
8. What is a residual in factor analysis? On what basis might it be judged "good" or "bad"?
9. When would you choose common factor analysis over PCA?